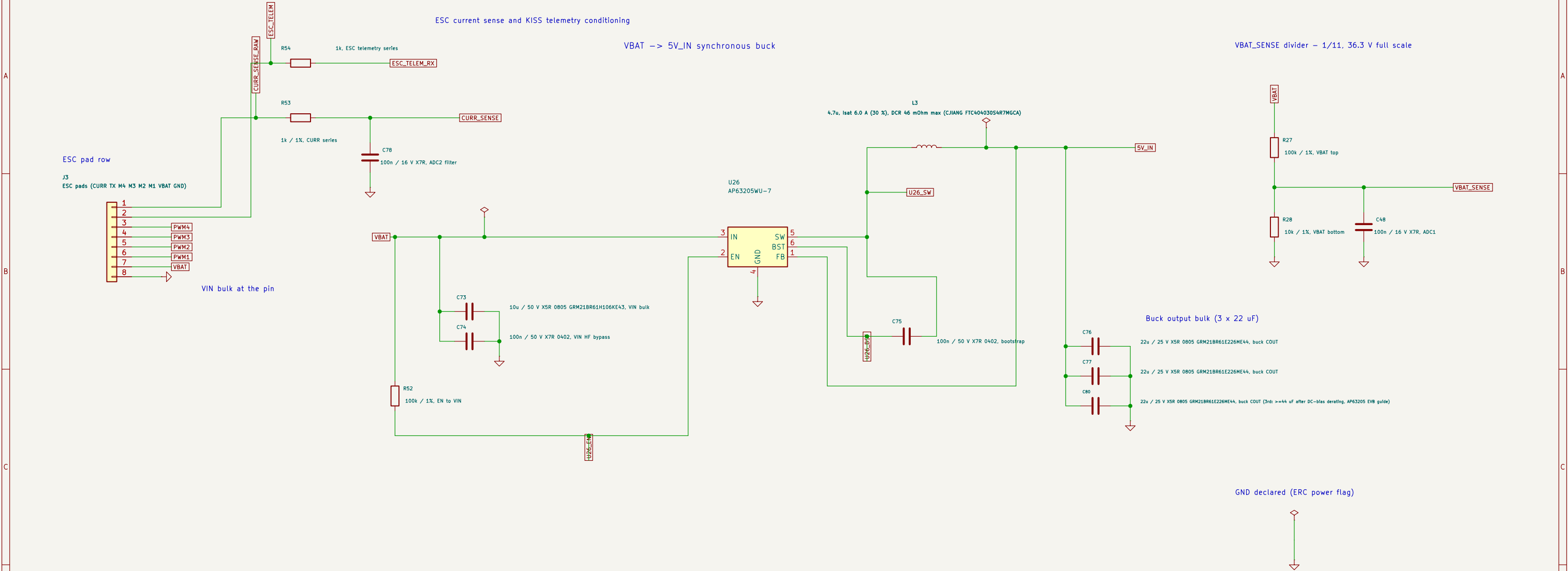


	1	2	3	4	5	6	7	8	
A	MARV V2 flight controller – schematic index. One page per subsystem; on each page the IC is drawn with its own decoupling, pull–ups, series resistors, test points and connectors wired to it, and a global label is used only where a net leaves the page. See DESIGN_SPEC.md for the design itself. Inputs: VBAT 6–25.2 V (2–6S) from the MicoAir AM32 ESC pad row J3, and USB VBUS. Rails: 5V_IN from the U26 AP63205 buck, V5_SYS from the U25 TPS2121 priority mux, V3V3_SYS from the TPS62913 buck, V3V3_ANA from the TPS7A20 LDO. Servo 5 V is V5_SYS, downstream of the mux.								
B	POWER 1 / ESC pad row, VBAT and the 5 V buck File: power_vbat.kicad_sch	BOARD 5 / USB–C, ESD, SWD pads File: usb_debug.kicad_sch	BOARD 3 / microSD socket File: sd.kicad_sch						
C	POWER 2 / TPS2121 priority power mux File: power_mux.kicad_sch	BOARD 2 / ICM–45686 six–axis IMU File: imu.kicad_sch	BOARD 3 / optional QSPI expansion socket (DNP) File: qspi_expansion.kicad_sch						
D	POWER 3 / 3.3 V system buck and analog LDO File: power_3v3.kicad_sch	BOARD 2 / BMP581 barometer File: baro.kicad_sch	BOARD 4 / IO array, status LED, mounting holes File: io_block.kicad_sch						
E	BOARD 1 / RP2354B File: mcu.kicad_sch	BOARD 2 / ADXL375 high–g accelerometer File: highg.kicad_sch							
F	Sheet: / File: MARV–V2.kicad_sch Title: MARV–V2 – schematic index Size: A3Date: 2026–09–16Rev: P0 – NOT FOR FAB KiCad E.D.A. 10.0.6Id: 1/12								
	1	2	3	4	5	6	7	8	

POWER 1 – ESC PAD ROW J3, VBAT INPUT, AP63205 5 V BUCK, ANALOG SENSE.
VBAT 6–25.2 V (2–6S) arrives dn J3 pin 7; U26 makes 5V_IN from it.



BOARD INPUTS: VBAT 6–25.2 V (2–6S) from the ESC pad row J3, and USB VBUS –5 V (J4). U26 (AP63205WU) makes 5V_IN from VBAT; 5V_IN is the priority input of the U25 power mux and also the servo 5 V bus.

ESC PAD ROW J3 (MicoAir AM32 4–in–1 ESC, FC–connection row). Order is the ESC silkscreen read left to right:
1 CURR – analog current sense from the ESC, 12.75 mV/A, 0–3.3 V -> R53/C78 -> GPIO42 (ADC2) on BOARD 1.
2 TX – KISS ESC telemetry, 3.3 V, 115200 baud, ESC output only -> R54 -> GPIO32 (PIO UART RX, GPIOBASE 16 window).
3-6 M4 M3 M2 M1 – 3.3 V DShot/PWM inputs via PIO (GPIOBASE 16 window), straight from the RP2354B (PWM4..PWM1 = GPIO103, 158, 39, 43).
7 VBAT – raw pack, 6–25.2 V (2–6S), the only board power input besides USB. 8 GND.
PITCH IS AN ASSUMPTION: MicoAir publishes no pad drawing, so MARV_Packages:PadRow_1x08_P2.00mm assumes 2.00 mm. The ORDER is authoritative: confirm the pitch against the physical ESC before fab.
The ESC–side low–ESR electrolytic on VBAT is MANDATORY: this board carries ceramic input capacitance only (C73 10 uF + C74 100 nF), and a ceramic–only pack connection rings to roughly twice pack voltage on hot plug. There is no TVS on VBAT by design – nothing that clamps below 32 V survives a 6S pack.

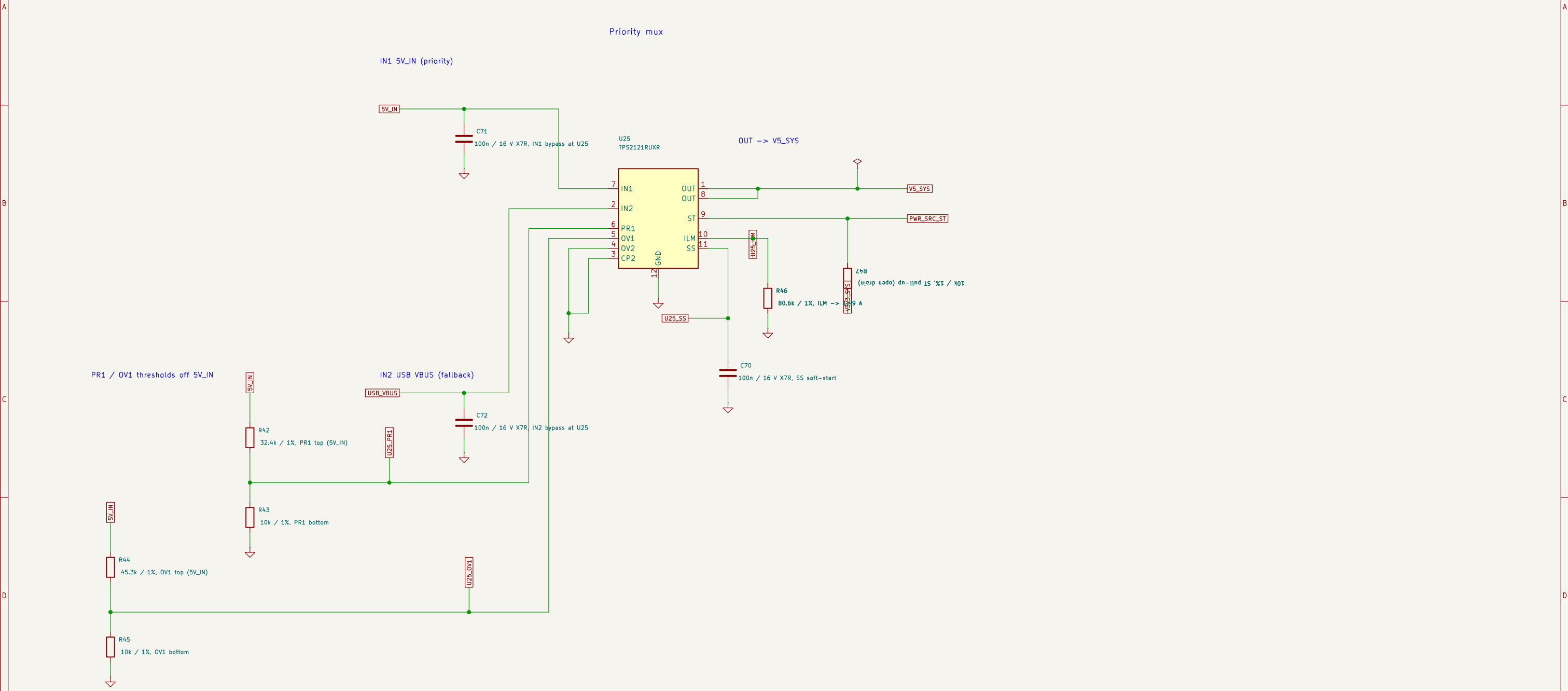
TELEMETRY AND ANALOG INPUTS (100 nF at each ADC pin).
VBAT_SENSE: R27/R28 100k/10k scale VBAT by 1/11, so the 3.3 V ADC full scale is 36.3 V. A 6S pack at its 22.2 V maximum reads 2.29 V and a 2S pack at 6.0 V reads 0.55 V, so the whole 2–6S window is on–scale with headroom; the divider draws 229 uA at 25.2 V. The old 5V_IN divider is gone with the 5 V input itself.
CURR_SENSE: the ESC current–sense output (J3 pin 1, 12.75 mV/A, 0–3.3 V) through R53 1k into C78 100 nF at the pin – a 100 us RC anti–alias/ESD network, well under the RP2350 ADC source–impedance guidance.
ESC_TELEM: the ESC KISS telemetry output (J3 pin 2, 3.3 V, 115200 baud, one–wire from the ESC) through R54 1k into GPIO32, read by a PIO UART (GPIOBASE 16 window). The resistor is series protection only, not a level shifter: the ESC pad row is 3.3 V CMOS on both the telemetry and the four motor lines.
VBUS_SENSE keeps 10k/15k, now on GPIO40 (ADC0); VBAT_SENSE moved to GPIO41 (ADC1).
D20 WS2812C–2020: VDD on V5_SYS with C79 100 nF, DIN from GPIO3 through R55 100 R. DOUT no–connect (single pixel). Its DIN VIH minimum is 2.7 V (datasheet Electrical Characteristics), which 3.3 V CMOS clears – that is why the C variant is used rather than a S050 WS2812B, whose VIH is 0.7 x VDD = 3.5 V.
QSPL* are the dedicated QSPL pads: they reach the In–package flash die AND the package pins (RP2350 datasheet Sec 14.3), so U24 shares the bus with GPIO0/QMI CS1n (FLASH_CS1) as its select.
RP2354 requires QSPLIOVDD = 3.3 V, and IOVDD = 3.3 V to run a second QSPL device (Sec 14.9).

U26 AP63205WU–7 VBAT BUCK (Diodes DS41326 Rev. 2–2), 3.8–32 V in, FIXED 5.0 V out (4.95/5.00/5.05 V, Electrical Characteristics VFB row), 2 A, 1.1 MHz with +/-0% frequency spread spectrum, 4 ms internal soft–start, 22 uA quiescent, TSOT–23–6, VIN abs max 35 V DC / 40 V for 400 ms vs a 25.2 V 6S pack. Component values are Table 3 “Recommended Component Selections for AP63205” and Figure 21 verbatim: L 4.7 uH (L3), C1 10 uF (C73), C2 3 x 22 uF (C76/C77/C80; datasheet minimum is 2 x 22 uF), C3 100 nF bootstrap BST–SW (C75).

FB (pin 1) is a SENSE input on the fixed–output parts and goes straight to the 5 V output – Sec 9 “Setting the Output Voltage”: only the adjustable AP63200/AP63201 take a divider. FB abs max 6.0 V.
EN (pin 2): “The EN pin is a high voltage pin and can be directly connected to VIN” (Sec 3 “Enable”), abs max 35 V, threshold 1.18 V rising / 1.10 V falling, internal 1.5 uA pull–up, R52 100k from VBAT is that connection with a series element, so a VBAT transient reaches the EN clamp through 100k rather than directly, and so a UVLO divider (Eq.1/Eq.2) or a start–delay cap (Eq.3) can be added later without touching the VBAT copper. Device UVLO is 3.5 V typ rising regardless.
L3: CJIANG FTC40403054R7MGCA, 4.7 uH +/-20% at 1 MHz / 1.0 Vrms, DCR 41 mOhm typ / 46 max, Irms 4.3 A typ / 4.0 A worst case, Isat 7.0 A typ / 6.0 A worst case, 4.1 x 4.1 x 3.0 mm molded metal–composite, shielded (CJIANG FTC series datasheet Rev 7.0 2025/11/05). CRITERIA ARE THE DATASHEET’S NOTES 3 AND 4: Irms is the DC current for an approximate 40 C rise, Isat the DC current for an approximate 30% drop in L0; Note 7 makes the usable rating the LESSER of the two, so L3 is Irms–limited at 4.0 A. Isat(30%) = 6.0 A worst case clears the AP63205 high–side peak current limit at its 3.1 A maximum by 94%, so the inductor does not saturate even in a current–limit or hiccup event, and Irms 4.0 A clears the 2.39 A operating peak. DCR is inside the datasheet’s “less than 100 mOhm” guidance: 46 mOhm max costs ~0.16 V at 2 A against the XGL4030’s ~0.13 V.
IT REPLACES THE COILCRAFT XGL4030–472MEC, which is orderable at JLCPCB (C7159276) but had 357 pieces in stock at \$7.82 each – 15% of the per–board component cost standing on one reel (reports/jlc–audit.md section 6). The FTC part is a STRICT IMPROVEMENT on the number that sized this inductor: worst–case Isat 6.0 A against the XGL’s 3.2 A at 20% / 4.4 A at 30%. What it costs is DCR (31.5 -> 46 mOhm max) and AEC–Q200 grading – the CJIANG sheet tests to AEC–Q200 METHODS but does not claim the qualification, and carries the usual “not warranted for aircraft equipment” clause.
VOLTAGE: datasheet Note 8 gives a 20 V DC withstand. In the on–time L3 sees VIN – VOUT, which at a full 6S pack (25.2 V) is 20.2 V – AT the rating. Below 5S this is not close. FLAGGED IN DESIGN SPEC “Open Items”.
4 x 4 x 3.0 mm. THIS ONE STILL CANNOT SHRINK THE WAY L2 DID: L2 is 3.0 x 3.0 x 2.0 because the TPS62913 runs it at ~1 A, but L3 has to clear the AP63205 3.1 A high–side limit and carry a 2.39 A operating peak, and no 3 x 3 part at 4.7 uH holds DCR under 50 mOhm at that current – the nearest, FTC30302004R7MBCA (LCSC C48888332, Isat 4.0 A worst

Sheet: /POWER 1 / ESC pad row, VBAT and the 5 V buck/ File: power_vbat.kicad_sch			
Title: POWER 1 / ESC pad row, VBAT and the 5 V buck			
Size: A3	Date: 2026–09–16	Rev: P0 – REVIEW ONLY	
KiCad E.D.A. 10.0.6		Id: 2/12	

POWER 2 – U25 TPS2121 PRIORITY POWER MUX
IN1 = 5V_IN (priority, from the U26 buck), IN2 = USB VBUS (fallback), OUT = V5_SYS.



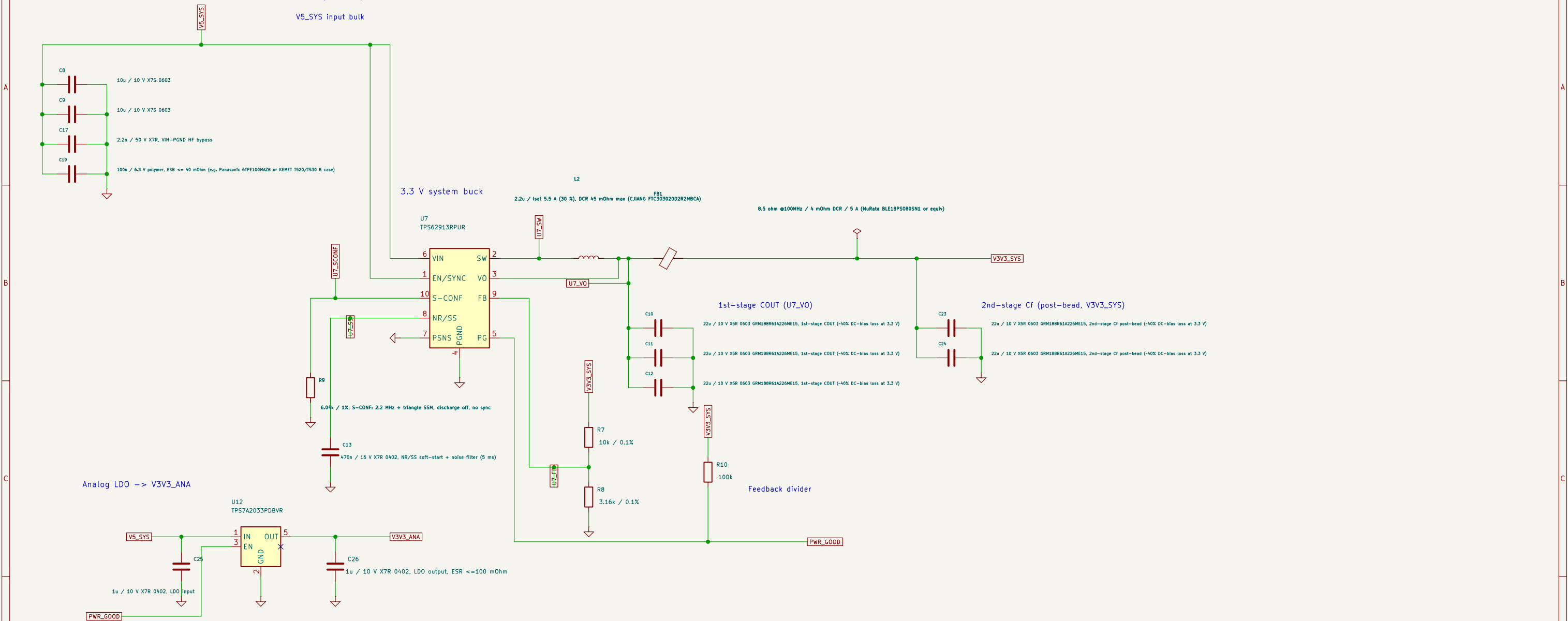
U25 TPS2121 PRIORITY POWER MUX (TI SLVSEA3F). IN1 = 5V_IN is the priority source, IN2 = USB_VBUS the fallback; OUT = V5_SYS.
Reverse-current blocking is always on for BOTH channels (Sec 9.3.6, IRCB 0.2/1/2 A, tRCB 10 us, VRCB 0/25/50 mV), so neither input can be back-fed from V5_SYS or from the other input. RON 56 mOhm typ / 100 mOhm max over temperature (Sec 7.5, VINx >= 5 V).
PR1 (Sec 10.2.4.1 Eq.5, VPR1 = VIN1 x Rb/(Rt+Rb) compared with VREF): R42 32.4k / R43 10k -> IN1 released when 5V_IN falls to 1.04 V / 0.23585 = 4.410 V typ and re-selected at 1.06 V / 0.23585 = 4.494 V typ (4.20–4.62 V / 4.28–4.66 V over the VREF spec).
The divider is centred on the RISING edge, not the falling one: the binding corner is a 5V_IN at its 4.75 V minimum meeting a VREF at its 1.10 V maximum, and 4.664 V worst-case rising keeps IN1 selected everywhere inside the declared 4.75–5.5 V window.
OV1 (Sec 9.3.5 / 10.2.4.2 Eq.6, same divider form): R44 45.3k / R45 10k -> IN1 rejected at 1.06 V / 0.180832 = 5.862 V typ (5.585–6.083 V over the VREF spec, so it can never trip inside the declared 4.75–5.5 V input window). OV2 is tied to GND: the USB channel has no overvoltage cut-off ("Connect to GND if not required", Sec 6). CP2 is tied to GND, which selects the Internal-VREF priority scheme of Table 9–3 – that is the mode PR1–vs–VREF describes, and it costs the TPS2121 fast-switchover path: the applicable spec is tSW = 100 us typ (Sec 7.5), not the 5 us tFSW that needs CP2 >= VREF.
ILM (Sec 9.3.2 Eq.2, ILM = 65.2 / RILM*0.861 with RILM in kOhm, valid 18–100 kOhm): R46 80.6k -> 1.49 A typ; the datasheet characterises RILM = 80k as 1.0 / 1.5 / 2.0 A min/typ/max. Fast-trip OCP is 2.4 x ILM (Sec 9.3.3).
SS (Sec 9.3.1 / Table 9–1): C70 100 nF -> 780 V/s at 5 V, so the ~120 uF on V5_SYS draws ~94 mA of inrush and OUT ramps in ~6.4 ms.
ST (Sec 6, Sec 10.2.3) is an open–drain status output: HIGH when IN1 (or neither input) drives OUT, LOW when IN2 does. R47 10k to V5V3_SYS is inside the RST = 6–20 kOhm recommended operating range of Sec 7.3; PWR_SRC_ST lands on RP2354B GPIO37 (pin 46).
Sec 11 gives no numeric minimum COUT, only "Increase the capacitance on OUT to avoid output voltage drop": C19 100 uF + C8/C9 2 x 10 uF on POWER 2 give 120 uF nominal, inside the 100–200 uF the datasheet design examples use, so no extra output capacitor is added here. C19 was a 220 uF / 10 V D case until the handover decks were re–run at 100 uF and every predicate still passed.
WHAT CHANGED WITH THE ON–BOARD BUCK: IN1 is no longer an external BEC of unknown quality, it is U26's regulated 5.00 V +/-1% output.
PR1 therefore sits far above its 4.494 V rising threshold whenever the pack is connected, so IN1 selection is unconditional in normal operation, and OV1 (5.862 V typ, 5.585 V worst case) can only trip on a U26 failure – it is now a backstop against a shorted high–side

FET pushing VBAT onto 5V_IN, not the mis–plug guard it used to be. The mis–plug case itself is gone: there is no 5 V input connector left to mis–plug, and VBAT does not reach V5_SYS by any path. The divider values are kept as they are because they still bound both ends of the window and cost nothing.

USB_VBUS feeds U25 IN2 directly: no inrush or current limiting on this sheet. Data ESD protection is on POWER 2. No servo rail connection.

Sheet: /POWER 2 / TPS2121 priority power mux/ File: power_mux.kicad_sch		
Title: POWER 2 / TPS2121 priority power mux		
Size: A3	Date: 2026–09–16	Rev: P0 – REVIEW ONLY
KiCad E.D.A. 10.0.6		Id: 3/12

POWER 3 – U7 TPS62913 BUCK (V5_SYS -> V3V3_SYS) AND U12 TPS7A20 LDO (-> V3V3_ANA)
AVIONICS ONLY: 300 mA continuous / 500 mA start peak, provisional, NO SERVO POWER.



V3V3_SYS feedback (TPS62913 datasheet Sec 8.2.2.2.6, Eq.8): $V_{OUT} = V_{FB} \times (1 + R1/R2)$; $V_{FB} = 0.8\text{ V typ. } 0.792\text{--}0.812\text{ V spec (Sec 6.5)}$.
 $R2 = 3.16\text{ k}\Omega\text{m} (\leq 5\text{ k}\Omega\text{m per datasheet noise guidance})$. $R1 = R2 \times (V_{OUT}/V_{FB} - 1) = 3.16\text{ k} \times 3.1625 = 9.99\text{ k}\Omega\text{m} \rightarrow \text{E96 } 10.0\text{ k}\Omega\text{m}$.
Actual $V_{OUT} = 0.8\text{ V} \times (1 + 10\text{k}/3.16\text{k}) = 0.8 \times 4.16456 = 3.3316\text{ V nominal, } +0.05\% \text{ off the } 3.33\text{ V the decks model}$.
THE PAIR IS 10k / 3.16k, NOT 15.8k / 4.99k, BECAUSE OF WHAT IS BUYABLE: 15.8 kOhm does not exist at 0.1 % in 0201 at JLCPCB at any stock level, so the divider as previously specified could not be built (reports/jlc-audit.md section 6).
Both halves are now Yageo RT0402BRD07 thin film, 0.1 % / 25 ppm, 0402: R7 = RT0402BRD0710KL (LCSC C190095).
R8 = RT0402BRD073K16L (LCSC C852759). SAME SERIES ON PURPOSE – a divider cares about the RATIO, and two parts from one thin-film series track each other far better than their individual 25 ppm/C tempcos suggest.
WINDOW, worst case, 0.1 % on BOTH resistors stacked with the VFB spec: ratio 10.01k/3.15684k = 3.17093 at one end and 9.99k/3.16316k = 3.15825 at the other, so V_{OUT} spans $0.792 \times 4.15825 = 3.293\text{ V}$ to $0.812 \times 4.17093 = 3.387\text{ V}$.
That is 3.29–3.39 V, not the 3.30–3.38 V the old text quoted – the old figure was the VFB term alone with the resistors treated as exact. Both windows sit inside the 3.135–3.60 V predicate the ngspice decks assert on V3V3_SYS.
Divider current rises from 160 uA (0.8 V / 4.99k) to 253 uA (0.8 V / 3.16k); 0.3 % of the 80 mA typical rail load, and the lower impedance is the direction the TPS62913 noise guidance points anyway.

S–CONF = 6.04 kOhm to GND (Table 7–1): 2.2 MHz switching, triangle spread–spectrum ON, output discharge OFF, no external sync.
2.2 MHz + 2.2uH matches the VIN=5V/VOUT<=3.3V design table (Table 8–2). TPS62913 runs fixed–frequency PWM at all loads, no light–load skip mode (Sec 7.4.1) – forced PWM is inherent; there is no separate MODE pin on this device.
VO (pin 3) senses the node between L1 and the ferrite bead (device internal loop); the FB divider senses V3V3_SYS after the bead for low–noise remote regulation (Sec 7.1 / 8.2.2.2.4).

V3V3_ANA feeds ICM–45686, BMP581, ADXL375 and the RP2354B ADC_AVDD pin (100 nF at each pin on the MCU sheet); VREG_AVDD is filtered from V3V3_SYS instead, so the analog rail carries no core–regulator current. U12 EN (pin 3) is on PWR_GOOD rather than V5_SYS.
The analog rail therefore starts only once the TPS62913 declares V3V3_SYS in regulation, and drops with it, which removes the window where the sensors were biased from an unregulated V5_SYS while the MCU was still held in reset.

PASSIVE PACKAGES (DESIGN_SPEC “Physical design”): resistors are 0402, ceramics <= 4.7 uF are 0402 and the 10–22 uF ceramics are 0603. RESISTORS ARE 0402, NOT 0201: JLCPCB has no Basic 0201 resistor of any value, so every 0201 line carried a per–unique–part Extended setup fee, and the 0.1 % values the U7 divider needs are not stocked in 0201 at all – see reports/jlc-audit.md, 0402 makes 8 of the 13 resistor values Basic and is hand–reworkable, which 0201 is not. C10/C11/C12 (1st–stage COUT) and C23/C24 (post–bead Cf) are 22 uF / 10 V X5R 0603 (GRM188R61A226ME15); at 3.3 V DC bias an X5R 22 uF 0603 keeps roughly 60 % of its nominal value, so the 66 uF nominal of the first stage is ~40 uF effective and the 44 uF post–bead stage is ~26 uF. The ngspice decks still carry the NOMINAL 66 uF (simulations/*.cir COUT), so the modelled rail is optimistic by that margin – an open item, not a result. Do not read the effective value off the BOM either.
The 22 uF 0603 ceramics remain machine–place parts – JLC assembles them, a bench iron will not rework one.
C19, THE V5_SYS BULK, IS 100 uF / 6.3 V POLYMER IN AN EIA–3528–21 (B) CASE, down from 220 uF / 10 V in an EIA–7343–31 (D) case: 8.9 x 4.9 mm of courtyard became 4.3 x 3.2 mm. The handover decks (simulations/source_handover.cir CPOLY, and the integrated deck) were re–run at 100 uF nominal / 80 uF derated and every predicate still passes – V5_SYS holds well above the 3.5 V floor across the 100 uS TPS2121 switchover – which is what licenses the smaller case. ESR must stay <= 40 mOhm (the value modelled); Panasonic 6TPE100MAZB or a KEMET T520/T530 B case. 6.3 V on a 5.0 V rail is a 1.26x derating, the normal polymer figure; a tantalum electrolytic would need 2x and would not qualify.

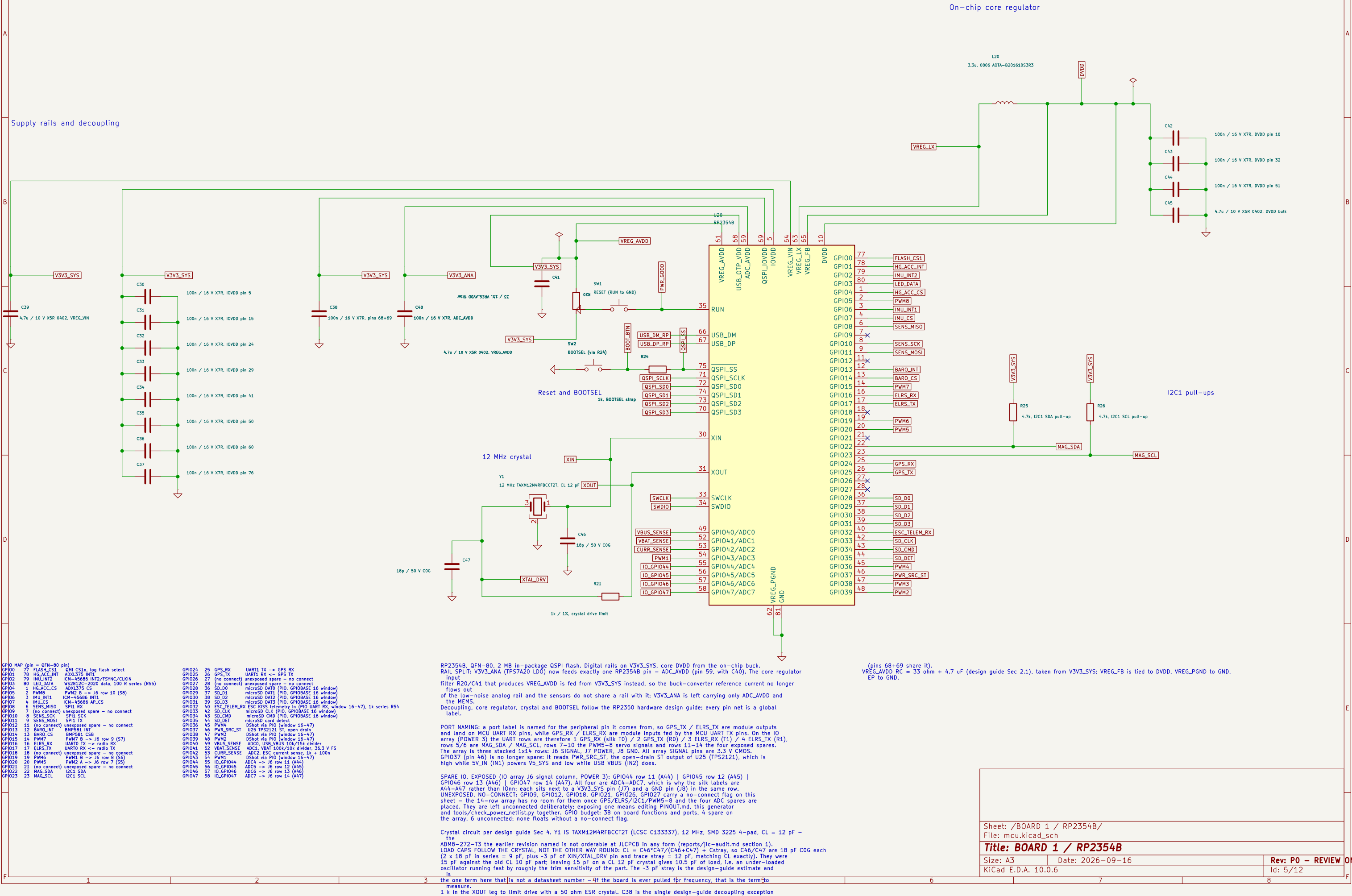
L2 INDUCTOR – CJIANG FTC303020D2R2MBCA, 3.0 x 3.0 x 2.0 mm molded metal–composite (SZ CJIANG FTC series datasheet Rev 7.0 2025/11/05, the sheet LCSC serves for C7423318): 2.2 uH +/-20% at 1 MHz / 1.0 Vrms, DCR 37 mOhm typ / 45 max. Irms 4.7 A typ / 4.3 A worst case, Isat 6.0 A typ / 5.5 A worst case. THE CRITERIA ARE THE DATASHEET’S OWN NOTE 3 AND 4.

Irms is the DC current for an approximate 40 C rise, Isat the DC current for an approximate 30% drop in L0. Note 7:

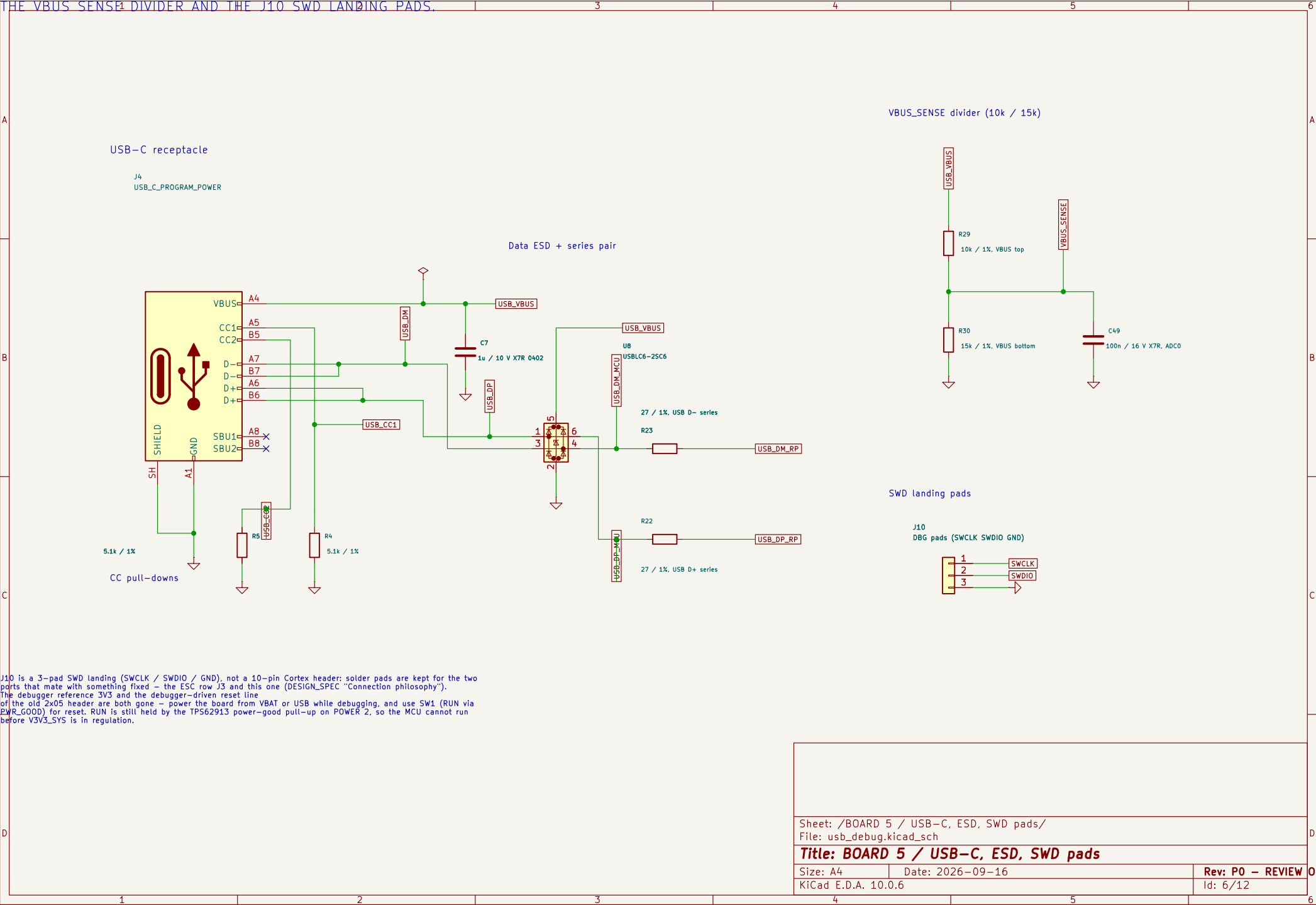
the usable rated current is the LESSER of the two, so this part is Irms–limited at 4.3 A, not saturation–limited. It replaces the Coilcraft XGL3020–222MEC, which is NOT ORDERABLE AT JLCPCB in any value (reports/jlc-audit.md section 6).
Same 3.0 x 3.0 x 2.0 envelope, same shielded (closed magnetic circuit) construction, DCR 30.5 typ / 36.5 max -> 37 typ / 45 max mOhm, which costs ~4 mW at the 0.8 A design load and is not a thermal factor.
OPERATING POINT: $dIL = V_{OUT} \times (1 - V_{OUT}/V_{IN}) / (L \times f_{sw}) = 3.3 \times 0.34 / (2.2\text{u} \times 2.2\text{M}) = 0.23\text{ A pk-pk nominal, } 0.29\text{ A at the } \sim 20\% \text{ inductance limit}$, so at the 0.8 A design load the peak inductor current is ~0.95 A and stays under ~1.05 A with VIN/fsw/L tolerances stacked. Isat(30%) = 5.5 A worst case is ~5.2x that.
THE FAULT CASE IS NOW COVERED, WHICH IT WAS NOT WITH THE XGL3020. The TPS62913 high–side current limit is ~4.5 A. The XGL3020 saturated at 2.85 A (30%), BELOW that limit, so a hard short on V3V3_SYS drove the old inductor into saturation before the IC limited, and the note here used to argue that away on soft–saturation and rail–size grounds. The FTC303020D saturates at 5.5 A worst case, ABOVE the ~4.5 A limit, so L2 now obeys the same rule as L3/U26: the IC limits first.
The argument that used to be needed is gone, not weakened – do not re–derive it.
RE–CHECK IF THE 3.3 V BUDGET EVER EXCEEDS ~3 A CONTINUOUS; the binding number then becomes Irms 4.3 A, not Isat. VOLTAGE: datasheet Note 8 gives a 20 V DC withstand. L2 sees VIN – VOUT = 1.7 V across it, two orders inside that. ORIENTATION: the FTC303020D is a symmetric two–terminal molded part with no start–lead marking, so unlike the XGL there is no preferred pad for the SW node – either way round is correct. Keep U7_SW on pad 1 anyway so the layout is unchanged.

Sheet: /POWER 3 / 3.3 V system buck and analog LDO/ File: power_3v3.kicad_sch		
Title: POWER 3 / 3.3 V system buck and analog LDO		
Size: A3	Date: 2026–09–16	Rev: P0 – REVIEW ONLY
KiCad E.D.A. 10.0.6		Id: 4/12

BOARD 1 – RP2354B, QFN–80, 2 MB IN–PACKAGE QSPI FLASH. Digital rails on V3V3_SYS, core DVDD from the on–chip buck. Supply pins fan out on the left into one decoupling group per rail.



BOARD 5 – J4 USB-C 2.0 RECEPTACLE, U8 USBLC6 DATA ESD, THE 27 R D+/D– SERIES PAIR,
THE VBUS SENSE DIVIDER AND THE J10 SWD LANDING PADS.



J10 is a 3-pad SWD landing (SWCLK / SWDIO / GND), not a 10-pin Cortex header: solder pads are kept for the two ports that mate with something fixed – the ESC row J3 and this one (DESIGN_SPEC "Connection philosophy"). The debugger reference 3V3 and the debugger-driven reset line of the old 2x05 header are both gone – power the board from VBAT or USB while debugging, and use SW1 (RUN via PWR_GOOD) for reset. RUN is still held by the TPS62913 power-good pull-up on POWER 2, so the MCU cannot run before V3V3_SYS is in regulation.

Sheet: /BOARD 5 / USB-C, ESD, SWD pads/
File: usb_debug.kicad_sch

Title: BOARD 5 / USB-C, ESD, SWD pads

Size: A4

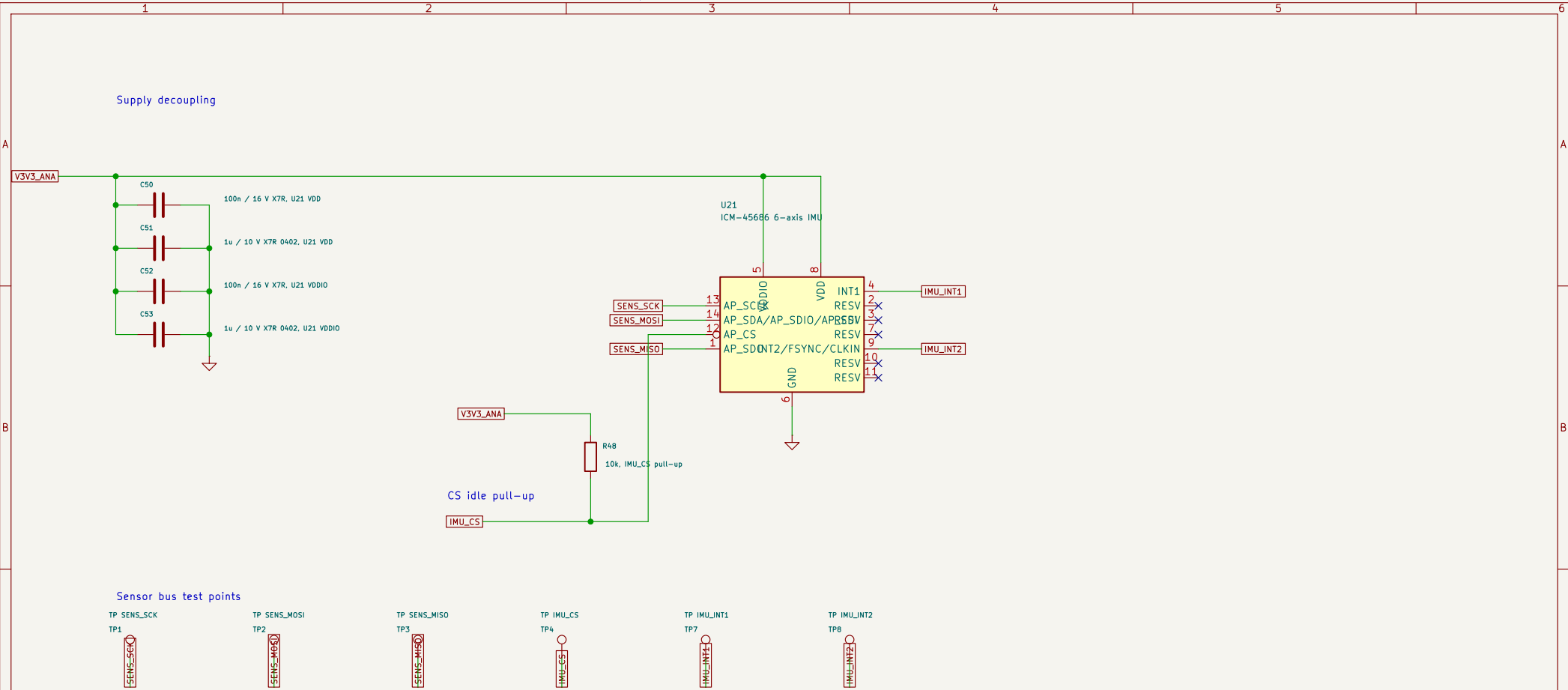
Date: 2026-09-16

Rev: P0 – REVIEW ONLY

KiCad E.D.A. 10.0.6

Id: 6/12

BOARD 2 – U21 ICM–45686 6–AXIS IMU on the shared sensor SPI bus, powered from V3V3_ANA.



All three sensors run 4–wire SPI on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus with one chip select each, and are powered from V3V3_ANA (TP57A20 LDO). Every supply pin on this sheet gets 100 nF + 1 uF; BMP581 and ADXL375 because their own datasheets call for the pair, the ICM–45686 as a deliberate addition – its datasheet BOM (DS–000577 Rev 1.0 Table 11) lists 0.1 uF only.

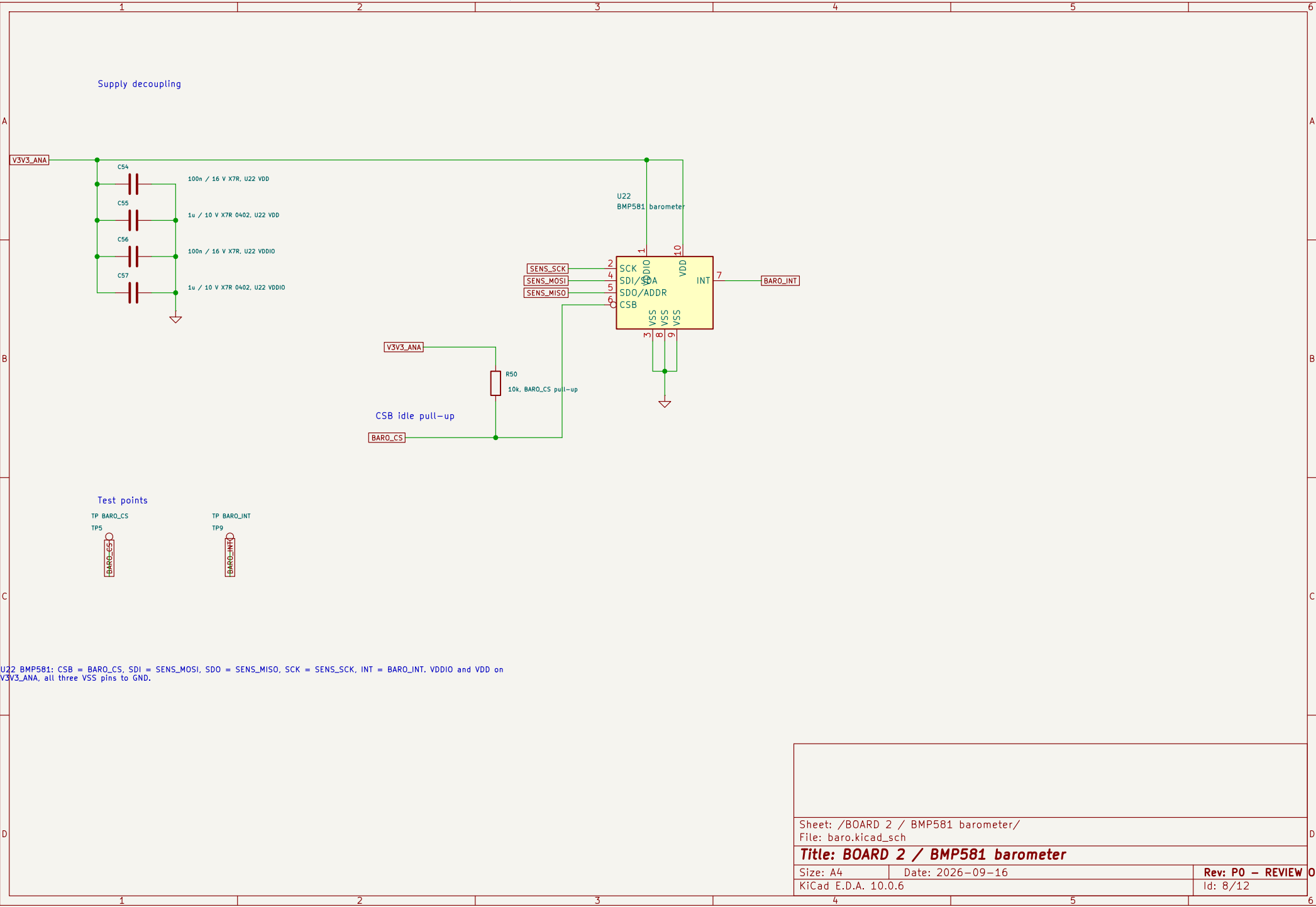
U21 ICM–45686 (TDK InvenSense DS–000577 Rev 1.0, Table 10 “Signal Descriptions” / Figure 4 pin–out, Single Interface SPI mode, Figure 10): AP_CS (pin 12) = IMU_CS, AP_SCLK (13) = SENS_SCK, AP_SDA/AP_SDIO/AP_SDI (14) = SENS_MOSI, AP_SDO/AP_ADD (1) = SENS_MISO, INT1 (4) = IMU_INT1, INT2/FSYNC/CLKIN (9) = IMU_INT2 – typed bidirectional because FSYNC/CLKIN are host–driven inputs while INT2 is a chip output; only the INT2 function is used here, VDD (8) and VDDIO (5) are both on V3V3_ANA; GND (6) is the only ground pin. Pins 2, 3, 7, 10 and 11 (RESV) are all left No Connect; Table 10 allows No Connect / VDDIO / GND for every RESV pin, and the Single–Interface SPI typical application schematic (Figure 10) shows all five left unterminated (internal pull enabled by default, per the per–pin notes in Table 10); they are only needed for the AUX1 or I2C–master dual–interface modes, neither of which this board uses. C50 (VDD) and C52 (VDDIO) are the 100 nF X7R of Table 11 “Bill of Materials” (C1/C2 in the datasheet); C51 (VDD) and C53 (VDDIO) add 1 uF on top of them. The datasheet BOM lists 0.1 uF only; this board keeps 100 nF + 1 uF per supply pin for consistency with U22/U23 and common flight–controller practice.

R48, R50, R51: 10k pull–ups to V3V3_ANA on the three SPI chip selects (IMU_CS, BARO_CS, HG_ACC_CS). R49 (the former second IMU chip–select pull–up) is removed along with that net – the ICM–45686 has a single chip select. Same rationale as R35 on FLASH_CS1 (BOARD 3): every sensor is held deselected before the RP2354B drives the pin, so a GPIO left floating through reset, BOOTSEL or a debugger halt cannot open a transaction on the shared SENS_SCK / SENS_MOSI / SENS_MISO bus. They bias to V3V3_ANA, the same rail as every VDDIO pin on this sheet, so there is no pull–up current path into an unpowered I/O supply and no CS pin is pulled above its own VDDIO during rail sequencing.

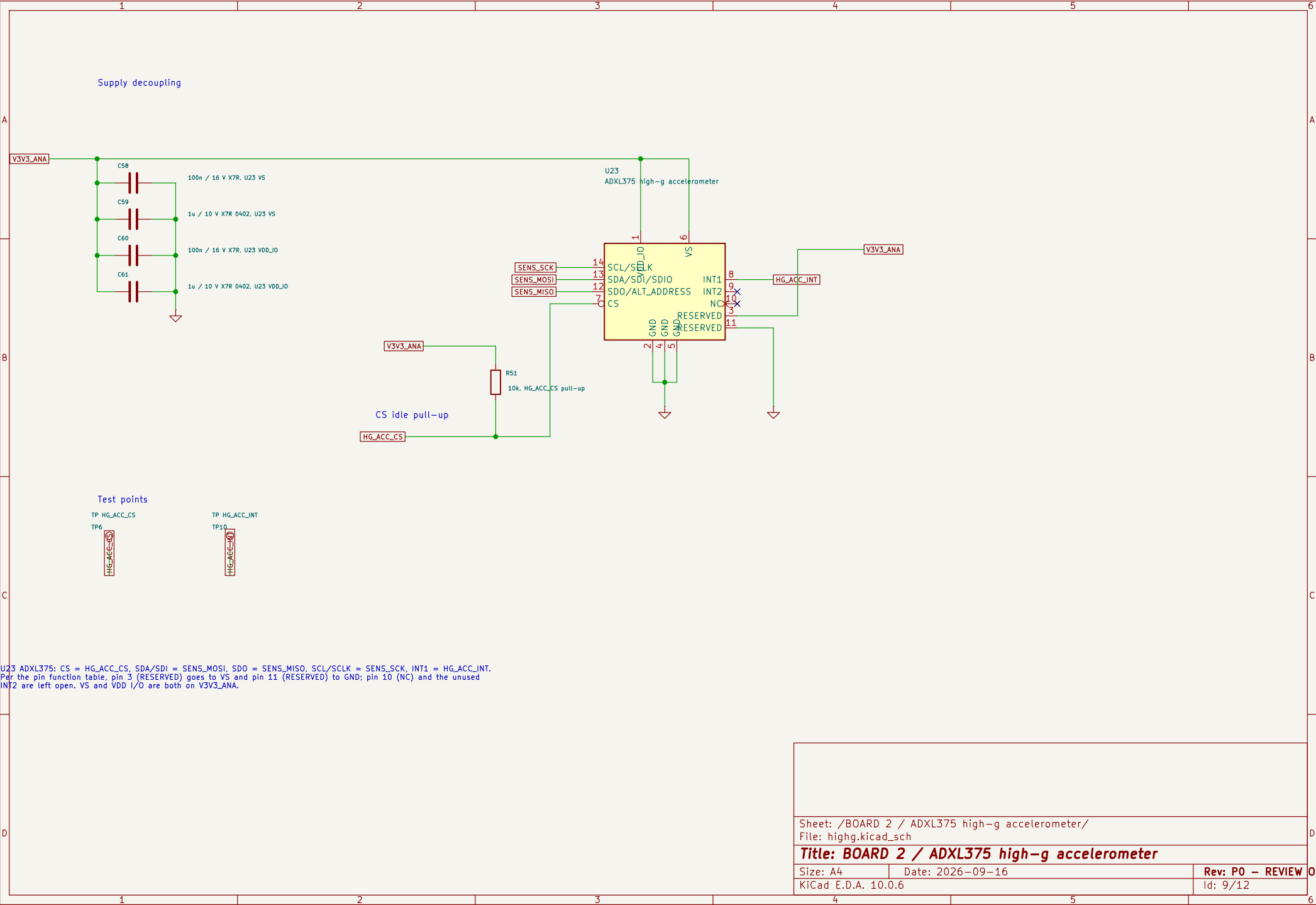
TEST POINTS TP1–TP10 (TestPoint:TestPoint_Pad_1.0x1.0mm, 1 x 1 mm bare F.Cu pad, no 3D model by design – tools/audit_footprints.py --no-3d-ok covers TestPoint_ along with the pad rows and mounting holes). TP1 SENS_SCK | TP2 SENS_MOSI | TP3 SENS_MISO | TP4 IMU_CS | TP5 BARO_CS | TP6 HG_ACC_CS | TP7 IMU_INT1 | TP8 IMU_INT2 | TP9 BARO_INT | TP10 HG_ACC_INT. Placement: inside the sensor cluster, within 6 mm of U21/U22/U23 (tools/setup_pcb.py checks it). They add ten stubs to the SPI bus, which is why they are 1 mm pads next to the parts rather than a header somewhere else; at 10 MHz SCK a short stub is a capacitive load, a long one is a transmission line. No series resistors and no ground pad of their own – probe ground comes off a mounting hole or the J6 array.

Sheet: /BOARD 2 / ICM–45686 six–axis IMU/ File: imu.kicad_sch		
Title: BOARD 2 / ICM–45686 six–axis IMU		
Size: A4	Date: 2026–09–16	Rev: P0 – REVIEW ONLY
KiCad E.D.A. 10.0.6		Id: 7/12

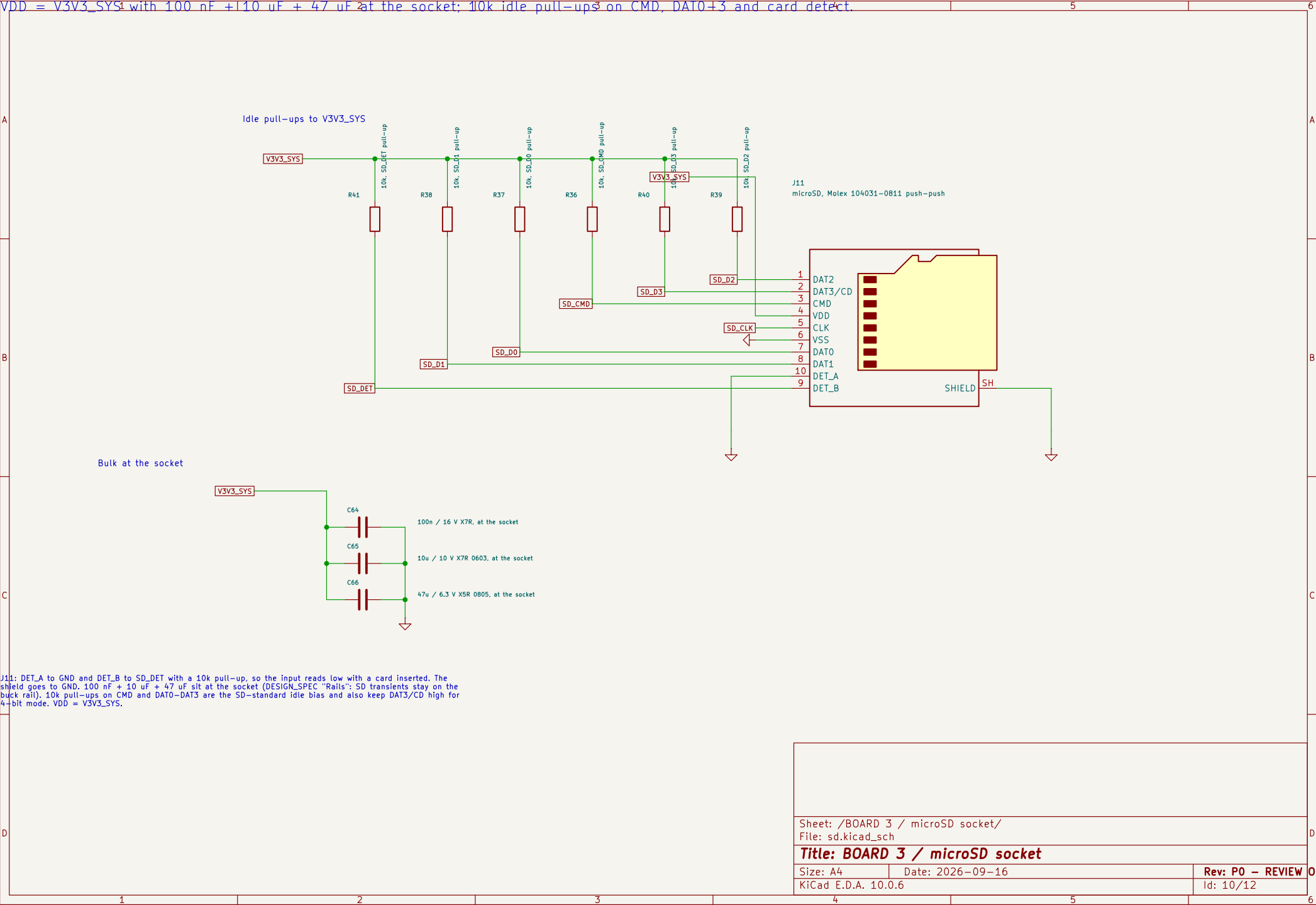
BOARD 2 – U22 BMP581 BAROMETER on the shared sensor SPI bus, powered from V3V3_ANA.



BOARD 2 – U23 ADXL375 HIGH-G ACCELEROMETER on the shared sensor SPI bus, from V3V3_ANA.

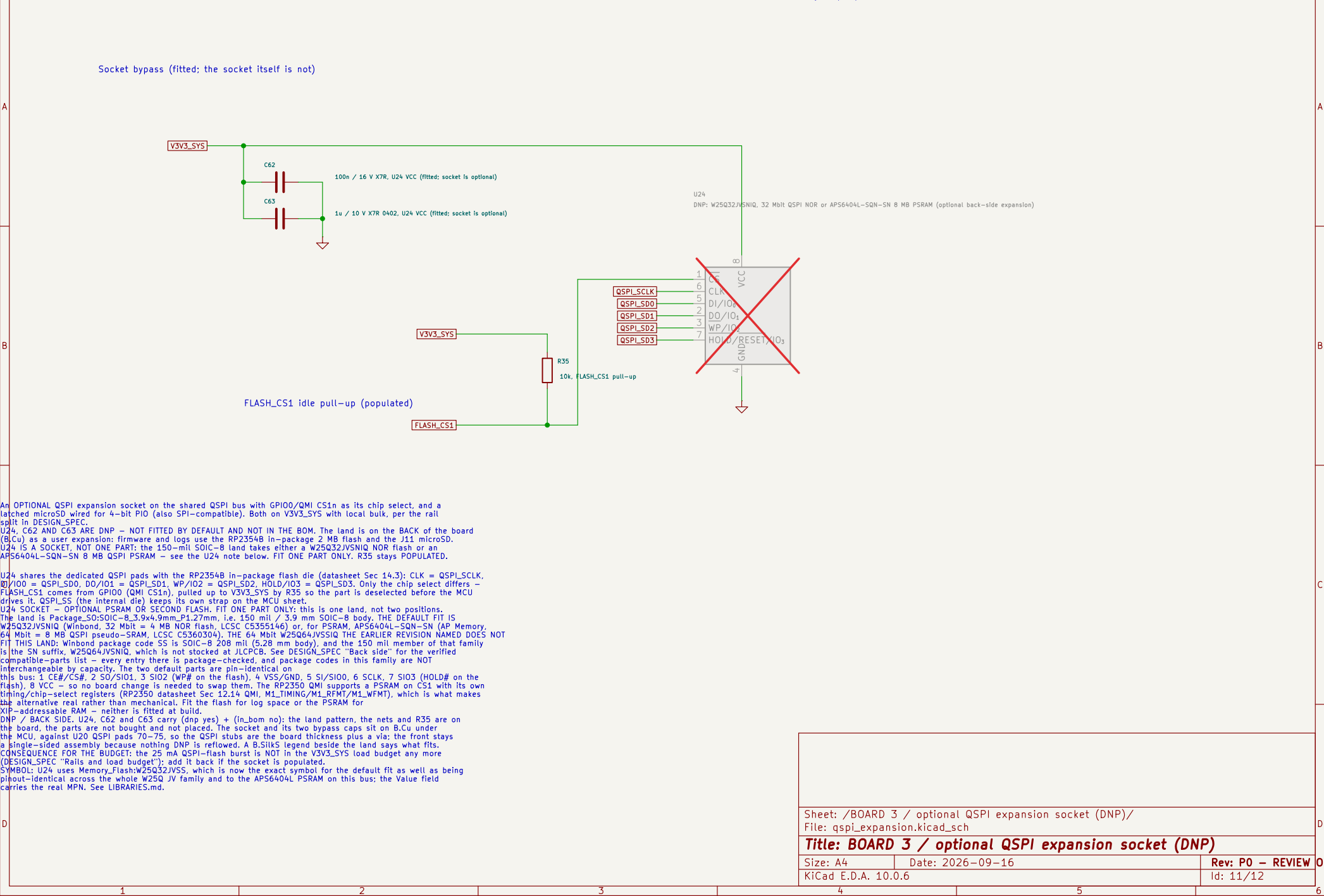


BOARD 3 – J11 MICROSD, LATCHED PUSH-PUSH, WIRED FOR 4-BIT PIO (also SPI-compatible).
VDD = V3V3_SYS with 100 nF + 10 uF + 47 uF at the socket; 10k idle pull-ups on CMD, DAT0+3 and card detect.



BOARD 3 – U24 OPTIONAL QSPI EXPANSION SOCKET, DNP, ON THE BACK (B.Cu).

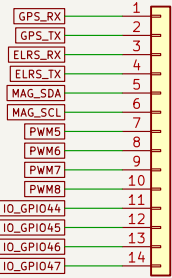
One 150-mil SOIC-8 land: fit EITHER a W25Q32JVSNQ NOR flash OR an APS6404L-SQN-SN PSRAM. R35 stays populated.



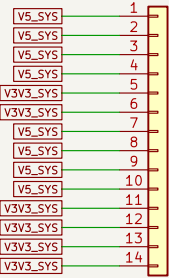
BOARD 4 – THE 3 x 14 IO BLOCK (J6 SIGNAL / J7 POWER / J8 GND), THE WS2812C STATUS LED AND THE FOUR M3 GROMMET HOLES. Every signal that leaves the board except the ESC row J3 and the DBG pads J10 leaves here.

IO array – one row per module lead

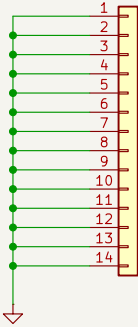
J6
IO array signal column (innermost)



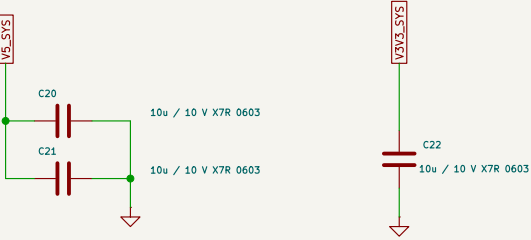
J7
IO array power column (5V / 3V3 per row)



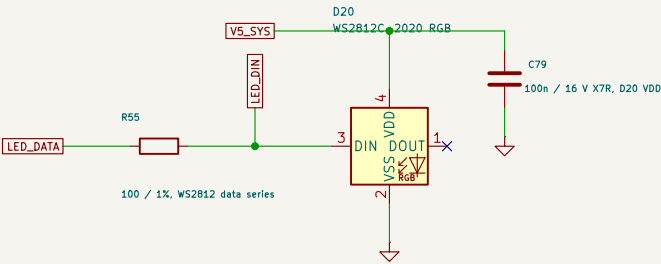
J8
IO array GND column (board edge)



Local bulk at the array



WS2812C status LED



Mechanical



M3 grommet hole, 4.0 mm NPTH



M3 grommet hole, 4.0 mm NPTH



M3 grommet hole, 4.0 mm NPTH



M3 grommet hole, 4.0 mm NPTH

IO ARRAY – ONE 3–COLUMN x 14–ROW THROUGH–HOLE BLOCK ON THE RIGHT EDGE, 2.54 mm, madflight FC3v2 style. It replaces the old 2x16 left–edge array AND the J12/J13/J14 servo block; every signal that leaves this board except the ESC row (J3) and the SWD pads (J10) now leaves on this one block. COLUMN RULE, from the board edge inward: GND (J8) | POWER (J7) | SIGNAL (J6). The SIGNAL column is the INNERMOST one, next to the MCU, so the 14 signal traces are the short ones; the two rails run down the outside where a plane stitch is cheap. A 3–wire module lead (signal, power, ground) plugs straight across ONE row. THREE Conn_01x14 SYMBOLS, not one 3x14; no 3–row generic connector symbol ships with this KiCad install, and three 1x14 rows with a body–trimmed courtyard about exactly on the 2.54 mm grid (see tools/setup_pcb.py). DS–009 NAMING IS KEPT: a port label is named for the PERIPHERAL pin it belongs to, so GPS_RX / ELRS_RX are module inputs driven by the RP2354B UART TX pins (rows 1 and 3, silk T0 / T1) and GPS_TX / ELRS_TX are module outputs that land on the MCU UART RX pins (rows 2 and 4, silk R0 / R1).

RAILS ON THE ARRAY: rows 1–4 (GPS, ELRS) and rows 7–10 (servos S5–S8) take V5_SYS; rows 5–6 (magnetometer) and rows 11–14 (the four exposed spares) take V3V3_SYS. Every row has its own GND pin in the J8 column, so a signal return is never more than 2.54 mm away and no row can be mis–plugged into a neighbour’s ground. SERVO 5 V NOW COMES FROM V5_SYS, NOT 5V_IN. The old J13 servo row tapped 5V_IN directly at the U26 buck output, ahead of the U25 priority mux, so servo current was outside the mux current limit. With one power column the array cannot carry two different 5 V nets under one “5V” silk label, so all eight 5 V rows are V5_SYS; servo current now crosses U25 and counts against its ILM limit (1.49 A, R46) and against the U26 buck. Budget servo current against BOTH – see DESIGN_SPEC “Rails and load budget”. CURRENT: rows 1–4 and 7–10 count against the U26 buck through the mux; rows 5–6 and 11–14 count in the V3V3_SYS budget. No fuse and no per–row current limit: the array is a system–integration connector, not a protected port.

MECHANICAL: H1–H4 are the four 4.0 mm NPTH grommet holes on the standard 30.5 x 30.5 mm pattern, centred on the board, each with a 5.0 mm copper keepout and an 8.0 mm top–side courtyard for the grommet flange. No pins, no nets. See DESIGN_SPEC “Physical design”.

IO ARRAY ROW MAP	(row n = J6 pin n signal J7 pin n power J8 pin n GND)
1 T0 GPS_RX	V5_SYS GND UART1 TX -> GPS RX (GPIO24)
2 R0 GPS_TX	V5_SYS GND UART1 RX <- GPS TX (GPIO25)
3 T1 ELRS_RX	V5_SYS GND UART0 TX <-> ELRS RX (GPIO16)
4 R1 ELRS_TX	V5_SYS GND UART0 RX <-> ELRS TX (GPIO17)
5 SDA MAG_SDA	V3V3_SYS GND I2C1 SDA (GPIO22), 4.7k pull–up on the MCU sheet
6 SCL MAG_SCL	V3V3_SYS GND I2C1 SCL (GPIO23), 4.7k pull–up on the MCU sheet
7 S5 PWM5	V5_SYS GND PWM2 A (GPIO20), servo 5
8 S6 PWM6	V5_SYS GND PWM1 B (GPIO19), servo 6
9 S7 PWM7	V5_SYS GND PWM7 B (GPIO15), servo 7
10 S8 PWM8	V5_SYS GND PWM2 B (GPIO5), servo 8
11 A44 IO_GPIO44	V3V3_SYS GND GPIO44 = ADC4, spare
12 A45 IO_GPIO45	V3V3_SYS GND GPIO45 = ADC5, spare
13 A46 IO_GPIO46	V3V3_SYS GND GPIO46 = ADC6, spare
14 A47 IO_GPIO47	V3V3_SYS GND GPIO47 = ADC7, spare

C20 / C21: V5_SYS local bulk at the array (rows 1–4 GPS/ELRS and rows 7–10 servo, the 5 V supply rows). C22: V3V3_SYS local bulk at the array (rows 5–6 and 11–14).

PWM MAP
PWM5 = GPIO20 (row 7, S5)
PWM6 = GPIO19 (row 8, S6)
PWM7 = GPIO15 (row 9, S7)
PWM8 = GPIO5 (row 10, S8)

ON THE ESC PAD ROW (J3, POWER 1)
PWM1 = GPIO43
PWM2 = GPIO39

PWM3 = GPIO38
PWM4 = GPIO36

Sheet: /BOARD 4 / IO array, status LED, mounting holes/
File: io_block.kicad_sch

Title: BOARD 4 / IO array, status LED, mounting holes

Size: A3 Date: 2026–09–16

Rev: P0 – REVIEW ONLY

KiCad E.D.A. 10.0.6

Id: 12/12